



# Identification and expression pattern analysis of *BpGRAS* gene family in *Bergenia purpurascens* and functional characterization of *BpGRAS9* in salt tolerance

Xin Lyu<sup>1</sup> · Nuomei Xu<sup>1</sup> · Jie Chen<sup>1</sup> · Wenqing Wang<sup>1</sup> · Feiyang Yan<sup>1</sup> · Zongxiang Jiang<sup>1</sup> · Qiankun Zhu<sup>1,2</sup> 

Received: 28 November 2024 / Accepted: 28 January 2025 / Published online: 13 February 2025  
© The Author(s), under exclusive licence to Springer Nature B.V. 2025

## Abstract

*Bergenia purpurascens* is an important medicinal, edible, and ornamental plant. It generally grows in extreme environments with complex stresses. The GRAS transcription factors play a crucial role in regulating plant stress tolerance and growth-development. There is no research on GRAS transcription factors in *B. purpurascens*. In this study, 29 *B. purpurascens* GRAS (*BpGRAS*) genes were identified based on *B. purpurascens* transcriptome data. These *BpGRAS* genes were classified into seven subfamilies according to phylogenetic analysis, while *BpGRAS1* was not classified into any other subfamilies. The motif analysis showed that the protein motifs in the same subfamily were relatively conserved. The expression pattern analysis of *BpGRAS* genes in different tissues and under salt stress showed that eight *BpGRAS* genes were differentially expressed under salt stress. The expression profiles showed that *BpGRAS9* might play an important role in salt response and the transgenic *Arabidopsis thaliana* lines with overexpressed *BpGRAS9* showed the enhanced salt tolerance. Root length and fresh weight were significantly increased in transgenic lines under salt conditions. The studies enhanced our comprehension of the function of *BpGRAS* and established a more foundation for exploring the molecular mechanisms underlying plant salt tolerance.

## Key message

This appears to be the first study to identify the GRAS genes in *B. purpurascens*. Their expression patterns under salt stress and the functional characterization of *BpGRAS9* provided more foundation for improving the plant salt tolerance.

**Keywords** GRAS gene family · *B. purpurascens* · Salt stress · *BpGRAS9*

## Introduction

*Bergenia purpurascens* is an evergreen, perennial, and stress-tolerant herb belonging to the genus *Bergenia* of the family Saxifrage, which is mainly distributed in Central and East Asia, and in China mainly in the southwestern plateau area (Zhang et al. 2017). *B. purpurascens* contains many

valuable bioactive compounds, such as Bergenin, Arbutin, and Catechin (Koul et al. 2020). Among them, bergenin is a natural product with a variety of benefits for human health, and is widely used clinically in the treatment of chronic bronchitis and tuberculosis (Yan et al. 2024). It also has important roles as an anti-arthritic, anti-diabetic and wound-healing agent (Chen et al. 2020a, b). In addition, bergenin showed potent anti-melanogenesis and skin-whitening effects by targeting TYR and TRP-2 proteins (Liu et al. 2024). Soil salinisation is becoming a global agroecological security crisis, and salt tolerance can lead to disruptions in plant metabolism that can seriously affect plant growth and crop yields (Yin et al. 2024; Zi et al. 2024). *B. purpurascens* has high medicinal values, and the extinction of wild *B. purpurascens* has only been prevented in recent years by artificial cultivation (Liu et al. 2023). The *B. purpurascens* planting area is relatively small. Natural factors, irrational

✉ Qiankun Zhu  
zhuqkswjtu@yeah.net

<sup>1</sup> Sichuan Engineering Research Center for Biomimetic Synthesis of Natural Drug, School of Life Science and Engineering, Southwest Jiaotong University, Chengdu 610031, China

<sup>2</sup> No. 111, North 1st Section, 2nd Ring Road, Chengdu 610031, China

irrigation methods, and the use of chemical fertilisers and pesticides accelerated the process of soil salinization, making cultivation of *B. purpurascens* difficult. Therefore, it is significant to study the salt-tolerance mechanism of *B. purpurascens* (Zhou et al. 2023).

GRAS transcription factors are a family of plant-specific transcription factors that play key roles in plant growth and development as well as in the response to adversity (Jaiswal et al. 2022). The GRAS gene family derives its name from the three earliest identified members of the family, gibberellin-insensitive (GAI), repressor of GAI (RGA) and scarecrow (SCR) (Bolle 2004). In general, GRAS proteins contain a conserved C-terminal GRAS domain, with the exception of the DELLA subfamily, which has a conserved DELLA domain at the N-terminus, while other GRAS proteins with highly variable N-terminal domain. The C-terminal GRAS domain can also be divided into five motifs, leucine heptad repeats I (LHRI), VHIIID, leucine heptad repeats II (LHRII), PFYRE, and SAW (Pysh et al. 2002). VHIIID, PFYRE and SAW are short or discontinuous conserved motifs (Hakoshima 2018).

The GRAS gene family has been reported in some monocotyledonous and dicotyledonous plants (Jaiswal et al. 2022). However, the classification of different subfamilies of GRAS genes is still not standardized. Based on the identification and analysis of GRAS gene families in *Arabidopsis thaliana* and rice, the GRAS gene family can be classified into eight subfamilies, DELLA, SCL3, SCR, SHR, LISCL, PAT1, HAM, LAS (Tian 2004; Tian et al. 2004). They perform distinct roles in regulation of plant growth-development and stress response. The DELLA subfamily mainly acts as negative regulators to regulate gibberellin (GA) signalling, affecting seed germination and controlling stem and root growth (Lee et al. 2008). In the root endodermis, the *A. thaliana* SCL3 protein is a tissue-specific integrator of the GA pathway, acting as a positive regulator to integrate and maintain the functional GA pathway by attenuating DELLA deterrents in the root endodermis (Heo et al. 2011). The SHR and SCR form the SCR/SHR complex as a key regulator of root growth and asymmetric cell division in *A. thaliana* (Helariutta et al. 2000; Koizumi and Gallagher 2013). The LISCL subfamily plays an important role in lily anther microsporogenesis (Morohashi et al. 2003). Wheat responds significantly to high light stress after overexpression of the wheat SCL subfamily *TaSCL14* gene (Chen et al. 2015a, b). PAT1, SCL5 and SCL21 are three proteins of the PAT1 subfamily in *A. thaliana* that are involved in the phytochrome A (PhyA) signalling pathway as positive regulators (Bolle et al. 2000; Torres-Galea et al. 2013), SCL13 in the same subfamily acts as a positive regulator of red light signalling downstream of phytochrome B (PhyB) (Torres-Galea et al. 2006). PAT1 subfamily also plays a role

in stress tolerance and two members of rice PAT1 subfamily, OsCIGR1 and OsCIGR2, are resistant to infection by pathogenic bacteria (Day et al. 2003, 2004). Overexpression of the *VaPAT1* gene in *A. thaliana* enhances the tolerance of transgenic *A. thaliana* to cold, drought and salinity (Yuan et al. 2015). HAM genes of *Petunia* are involved in signalling from lateral organ primordia and stem pro-vessels and play a key role in maintaining shoot apical meristem (Stuurman et al. 2002). Two genes of the LAS subfamily in tomato and *A. thaliana*, *Ls* and *LAS*, are required for the formation of axillary meristematic tissues, and the homologue *MOCI* gene in rice regulates the tillering development (Greb et al. 2003; Schumacher et al. 1999).

Until recently, *B. purpurascens* GRAS (*BpGRAS*) gene family has not been reported. In this study, the members of the *BpGRAS* gene family were identified and subjected to expression pattern analysis based on our previously obtained transcriptome data of *B. purpurascens* (Zhang et al. 2023; Zhu et al. 2024). *BpGRAS9*, a member of PAT1 family, was shown to exhibit a significant response to salt stress. Consequently, *BpGRAS9* was isolated and subsequently overexpressed in *A. thaliana* to elucidate its function. The results indicated that *BpGRAS9* acts as a positive regulator in salt tolerance. This is the first study to identify the GRAS genes in *B. purpurascens*, their expression patterns under salt stress and the functional characterization of *BpGRAS9* provided a more foundation for improving the plant salt tolerance.

## Materials and methods

### Plant materials and growth conditions

*B. purpurascens* used in the study were procured from Wenchuan Botanical Garden, Wenchuan County, Sichuan Province, China, and planted in organic nutrient soil at an incubation temperature of 22 °C under long day conditions (16 h light, 70  $\mu\text{E m}^{-2} \text{s}^{-1}$  and 8 h dark). Species information was confirmed by DNA barcoding (ITS2 and psbA-trnH sequences) (Wei et al. 2021). *A. thaliana* (ecotype Columbia Col-0) was used for genetic transformation in this study, and transgenic lines OE2 and OE3 were used for subsequent studies.

### Data sources and identification of the *BpGRAS* gene family

The RNA-seq data of the three tissues (root, stem and leaf) of *B. purpurascens* and the salt-treated leaf were performed by our group (ncbi: PRJNA987807, PRJNA1177679). The protein sequence file of *B. purpurascens* was extracted from

the transcriptome data of *B. purpurascens* to be used as a local protein database, and the protein sequences of the *A. thaliana* GRAS gene family were downloaded from the TAIR database. BLASTP search was performed using TBtools with the (*A*) *thaliana* GRAS protein sequence as the query sequence and the (*B*) *purpurascens* protein sequence as the target sequence, E-value <  $10^{-5}$  (Chen et al. 2020a, b). At the same time, the pfam.a.hmm files were downloaded from the pfam database and the ‘simple hmmer’ search in TBtools was used to search the local protein database with GRASP-FID (pf03514) (Chen et al. 2020a, b). The two results were integrated together and redundancy was manually removed to obtain the *B. purpurascens* GRAS protein sequences. The obtained *B. purpurascens* GRAS protein sequences were further submitted to the nr database to remove the genes that do not contain the GRAS structural domains to obtain the final *B. purpurascens* GRAS family related genes.

Physicochemical properties such as molecular weight and isoelectric point of the BpGRAS family protein sequences were analysed using the online software ExPASy-ProtParam tool (<https://web.expasy.org/protparam/>). BpGRAS proteins were analysed for subcellular localisation using the online software Plant-mPLOC Server (<http://www.csbio.sjtu.edu.cn/bioinf/plant-multi/#>) (Wen et al. 2024; Zhang et al. 2024).

### Phylogenetic analysis and conserved motif analysis of the BpGRAS

The 29 GRAS protein sequences of *B. purpurascens*, 34 GRAS protein sequences of *A. thaliana* and 60 GRAS protein sequences of rice were imported into the MEGA7 software, and we used the Muscle program to perform multiple sequence comparison calculations, and the results of the calculations were used in the neighbour-joining method to construct the rootless phylogenetic tree (Edgar 2004). (*A*) *thaliana* GRAS protein sequence files downloaded from TAIR, rice GRAS protein sequence files downloaded from Phytozome v13 (Zhao et al. 2022). Phylogenetic trees were landscaped using the online software Evolview (<http://www.evolgenius.info/evolview/treeview>). The conserved motifs of (*B*) *purpurascens* GRAS family proteins were analyzed using the online software MEME Suite 5.1.0 (<http://meme-suite.org/tools/meme>), with the number of motifs set to 10 and the rest of the parameters at default parameter. The BpGRAS phylogenetic tree and protein conserved motifs were visualized using TBtools (Chen et al. 2020a, b).

### Analysis of the expression pattern of BpGRAS in different tissues and after salt treatment

RNA-seq data from three tissues of *B. purpurascens* roots, stems, and leaves, as well as salt-treated leaves, were used

to calculate gene expression levels via <http://deweylab.bios.tat.wisc.edu/rsem/> RSEM (v1.3.0.1) (Li and Dewey 2011), and expression levels of single genes were converted to TPM (transcripts per million) values. Differential expression analysis was performed by DESeq2, and differentially expressed genes between samples were screened using  $\log_2|\text{fold change}| \geq 1$  &  $\text{P}_{\text{adjust}} < 0.05$ . Gene TPM values were imported into TBtools to map expression.

### RNA extraction and RT-qPCR analysis

Total RNA was extracted from the first fully expanded leaf starting at the top of *B. purpurascens* control samples and salt-treated samples using the Trelief® RNAprep Pure Plant Plus Kit (Tsingke, China). RNA concentration and purity were detected using nano drop 2000 (Thermo Fisher Scientific, USA), and RNA integrity was detected by agarose gel electrophoresis. First-strand cDNA was reverse transcribed from RNA using the Evo M-MLV RT mix Kit and gDNA Clean for RT-qPCR Ver. 2 (Accurate Biology, China). RT-qPCR was performed on eight genes to validate the RNA-seq data. Normalization of expression data using 18 S rRNA as an internal reference (Yin et al. 2017). RT-qPCR was performed with SYBR GREEN I (Precision Bio, China) on an Analytik Jena qTOWER3 G fluorescent quantitative PCR instrument (Analytik Jena AG, Germany). Cycling conditions were 95 °C pre-denaturation for 30 s, 95 °C pre-denaturation for 5 s, and 60 °C pre-denaturation for 30 s for 40 cycles. Primers for RT-qPCR were designed by Primer-Blast and are shown in Table S1. Relative gene expression was calculated using the  $2^{-\Delta\Delta C_t}$  method.

### Vector construction and A. Thaliana transformation

The termination codon-free *BpGRAS9* gene fragment was amplified by PCR using specific primers containing *Xba*-I and *Stu*-I restriction sites (Table S2), and *BpGRAS9* was inserted into the pCAMBIA1300 vector by seamless cloning. The *BpGRAS9* overexpression vector was introduced into *Agrobacterium tumefaciens* GV3101, which was then transformed into *A. thaliana* by the floral dip method (Clough and Bent 1998). Wild-type (WT) and transgenic *A. thaliana* seeds were harvested and surface-sterilized with alcohol for 3 min, then surface-sterilized with sodium hypochlorite for 3 min, then washed three times with sterile water for one minute each, and the transgenic seedlings were screened on 1/2 MS medium containing hygromycin (25 mg/L) and timentin (25 mg/L) at an incubation temperature of 22 °C under long-day conditions (16 h light, 70  $\mu\text{E m}^{-2} \text{s}^{-1}$  and 8 h dark), resulting in six *BpGRAS9* overexpression lines (OE1–6). OE2 and OE3 with high expression were selected to be used for subsequent studies.

## Salt stress treatment and phenotype analysis of plants

*B. purpurascens* were poured into 100 ml of solution without and with 120 mM NaCl, and after 6 h the first fully expanded leaf starting from the top was taken and transferred to a test tube and immediately frozen in liquid nitrogen and placed at -80 °C for storage.

For fresh weight and root length analysis of *A. thaliana*, seeds were surface-sterilized and sown on 1/2MS medium without NaCl, stratified at 4 °C for 3 days and then transferred to a light incubator to grow 4 days, and then the seedlings with comparable root lengths were picked and transferred to 1/2MS medium without NaCl or with 120 mM NaCl for 8 days. For albino rate and chlorophyll content analysis of *A. thaliana*, seeds were sown on 1/2 MS medium without NaCl or with 120 mM NaCl, stratified at 4 °C for 3 days and then transferred to a light incubator for 10 days. The incubator was maintained at an incubation temperature of 22 °C under long-day conditions (16 h light, 70  $\mu\text{E m}^{-2} \text{s}^{-1}$  and 8 h dark). The Chlorophyll content of *Arabidopsis* control and transgenic plants was determined with reference to Jiang et al. (2013) (Jiang et al. 2013).

## Determination of ROS accumulation

To explore the accumulation of ROS, seeds were surface-sterilized and sown on 1/2MS medium without NaCl, stratified at 4 °C for 3 days and then transferred to a light incubator to grow 4 days, and then the seedlings with comparable root lengths were picked and transferred to 1/2MS medium without NaCl or with 120 mM NaCl for 3 days. All samples were collected and stained with freshly configured NBT and DAB solutions for 30 min (10 mg of NBT and DAB dissolved in 10 ml of phosphate buffer, respectively) and then observed under a microscope after complete decolourisation with 75% ethanol (Myouga et al. 2008). Determination of POD and CAT content was done by Peroxidase (POD) activity assay kit and Catalase (CAT) activity assay kit purchased by Sangon Biotech. Determination of proline content with reference to the method of Yuan (Yuan et al. 2015).

## Results

### Identification of the GRAS gene family in *B. purpurascens*

The obtained *B. purpurascens* protein sequences were imported into TBtools, and the *B. purpurascens* GRAS gene sequences were searched using ‘blastp’ and ‘simple

hammer’ in TBtools. The identified genes were removed from redundancy and identified as containing conserved GRAS domains on NCBI, and finally 29 *BpGRAS* genes were obtained, which were named *BpGRAS1-29* in order (Table S1).

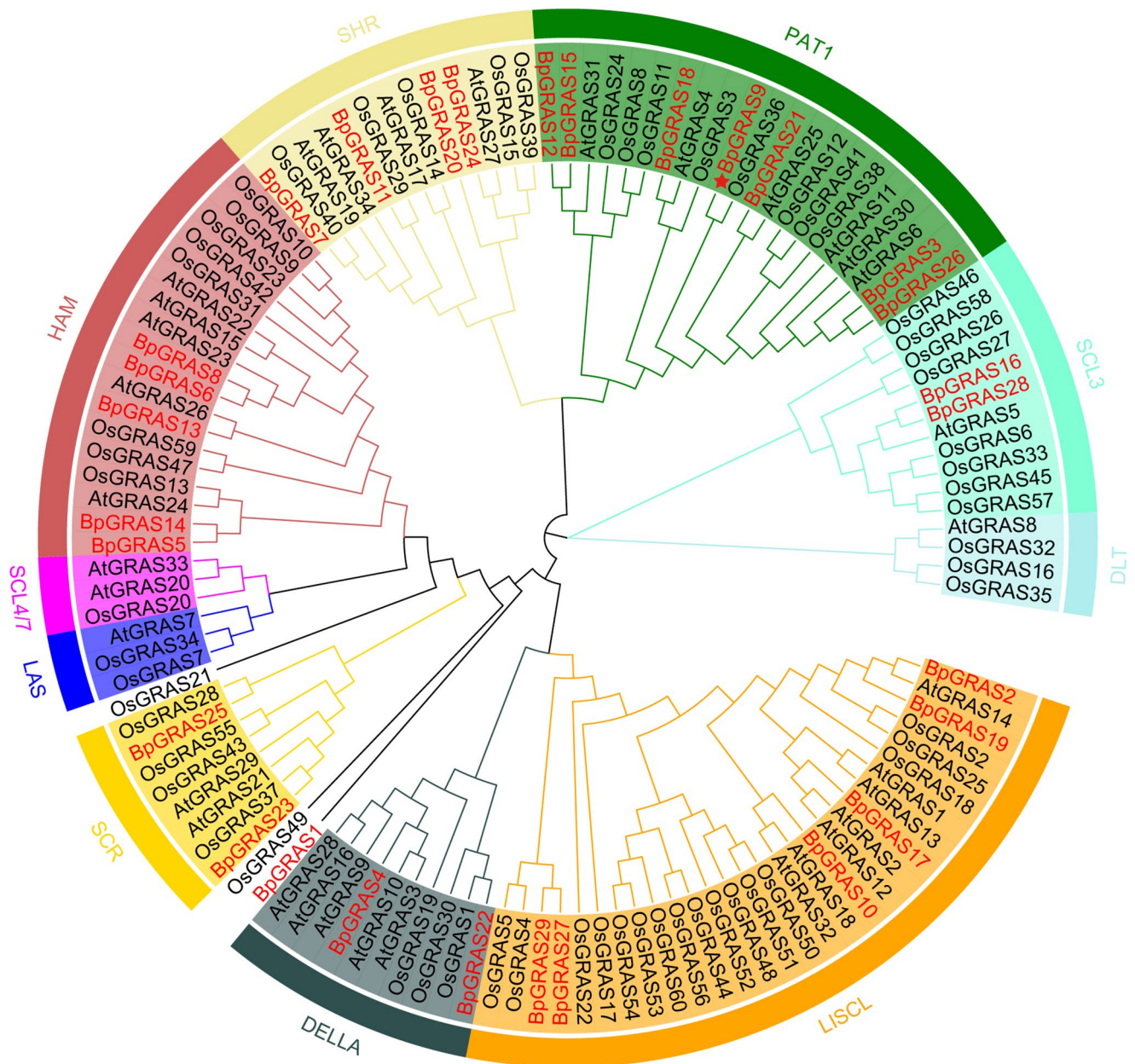
Analysis of the physicochemical properties of BpGRAS proteins showed that the number of amino acids (AA) and molecular weight (Mw) of these BpGRAS proteins varied significantly (Table S2). BpGRAS20 had the fewest amino acids, only 424 amino acids with a molecular weight of 47.66 kDa. BpGRAS17 had the most amino acids, with 750 amino acids and a molecular weight of 85.58 kDa. The mean isoelectric point (pI) was 5.7, and all the GRAS proteins were acidic with isoelectric points less than 7 except for BpGRAS2, which was basic with an isoelectric point greater than 7. BpGRAS3, BpGRAS18, BpGRAS5, and BpGRAS14 were predicted to be strongly acidic. The instability index (II) ranges from 41.97 to 59.83 and the predicted results indicate that all BpGRAS proteins are unstable. The aliphatic index (AI) ranged from 67.67 to 96.62, and all BpGRAS proteins were predicted to be hydrophilic (GRAVY) with a total mean value of less than 0 and considered to be hydrophilic. The subcellular localization prediction of BpGRAS proteins with Plant mPloc showed that all BpGRAS proteins are distributed in the nucleus, suggesting that they might function as transcription factors in the nucleus.

### Phylogenetic and motif composition analysis of BpGRAS proteins

A total of 123 GRAS proteins, including 29 BpGRAS proteins, 34 GRAS proteins from *A. thaliana*, and 60 GRAS proteins from rice, were subjected to phylogenetic analysis. A rootless phylogenetic tree was constructed (Fig. 1) by MEGA7 using the neighbour-joining method. The results showed that 29 BpGRAS proteins were assigned to seven subfamilies in (*A. thaliana*), and there were no homologous (*B. purpurascens*) GRAS members in the rice-specific subfamily. The distribution of BpGRAS members across the four subfamilies LISCL, PAT1, SHR, and HAM was relatively balanced, comprising 6, 7, 4, and 5 members, respectively, while the subfamilies SCL3, SCR, and DELLA each comprised only two BpGRAS members, and BpGRAS1 did not belong to any of the above subfamilies, and probably belonged to the *B. purpurascens* specific subfamily.

Ten conserved protein motifs of BpGRAS were identified using the MEME online platform (Table S3), which provided a reference for the functional resolution of the BpGRAS protein (Fig. 2). From the results, it could be seen that most of the conserved motifs were located in the C-terminal region. Among these 10 conserved protein motifs,



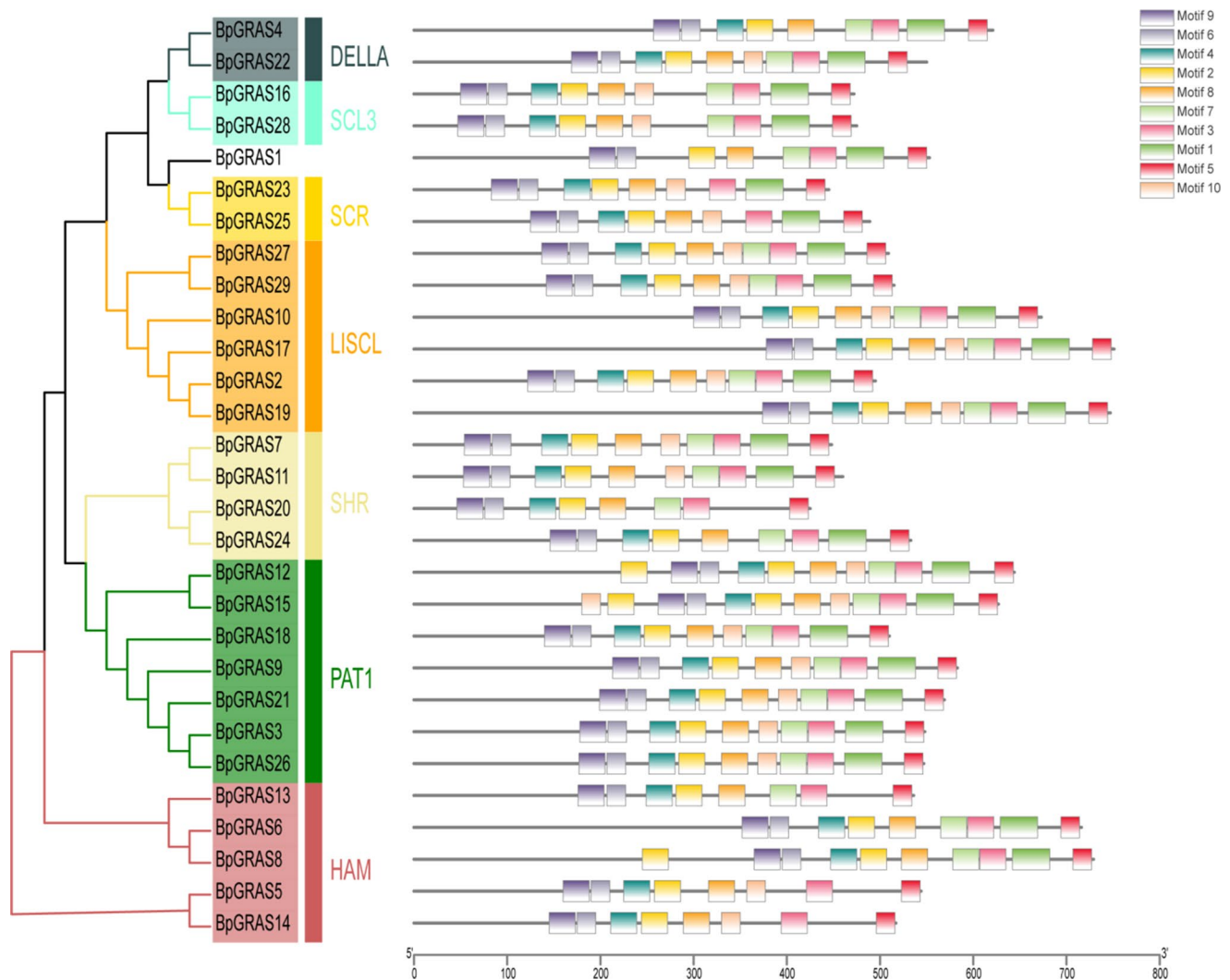


**Fig. 1** Phylogenetic tree of GRAS proteins from *B. purpurascens*, *A. thaliana*, and rice constructed by the neighbour-joining method. Different subfamilies are shown in different colours

motif 9, motif 6, motif 2, motif 8, motif 3, and motif 5 were highly conserved and were present in all BpGRAS proteins. Furthermore, variations in the conserved motifs of GRAS proteins across different subfamilies were observed, potentially explaining the functional disparities among them. No deletion of protein motifs occurred in the SCL3, LISCL, and PAT1 subfamilies. BpGRAS1 was the sole protein lacking both motif 4 and motif 10, which might account for its exclusion from the aforementioned eight subfamilies.

### The expression pattern of BpGRAS genes in different tissues

The expression levels of *BpGRAS* genes across different tissues (Table S4) from the RNA-seq data of roots, stems and leaves of *B. purpurascens* established a foundation for investigating the role of *BpGRAS* gene family in the growth and development of *B. purpurascens*. The TPM (transcripts per million) values of *BpGRAS* genes were calculated by RSEM (v1.3.1, <http://deweylab.biostat.wisc.edu/rsem/>) and plotted as a heatmap by TBtools to show the expression



**Fig. 2** Distribution of *B. purpurascens* GRAS protein motifs, different motifs are indicated by different colours, motifs 1–10

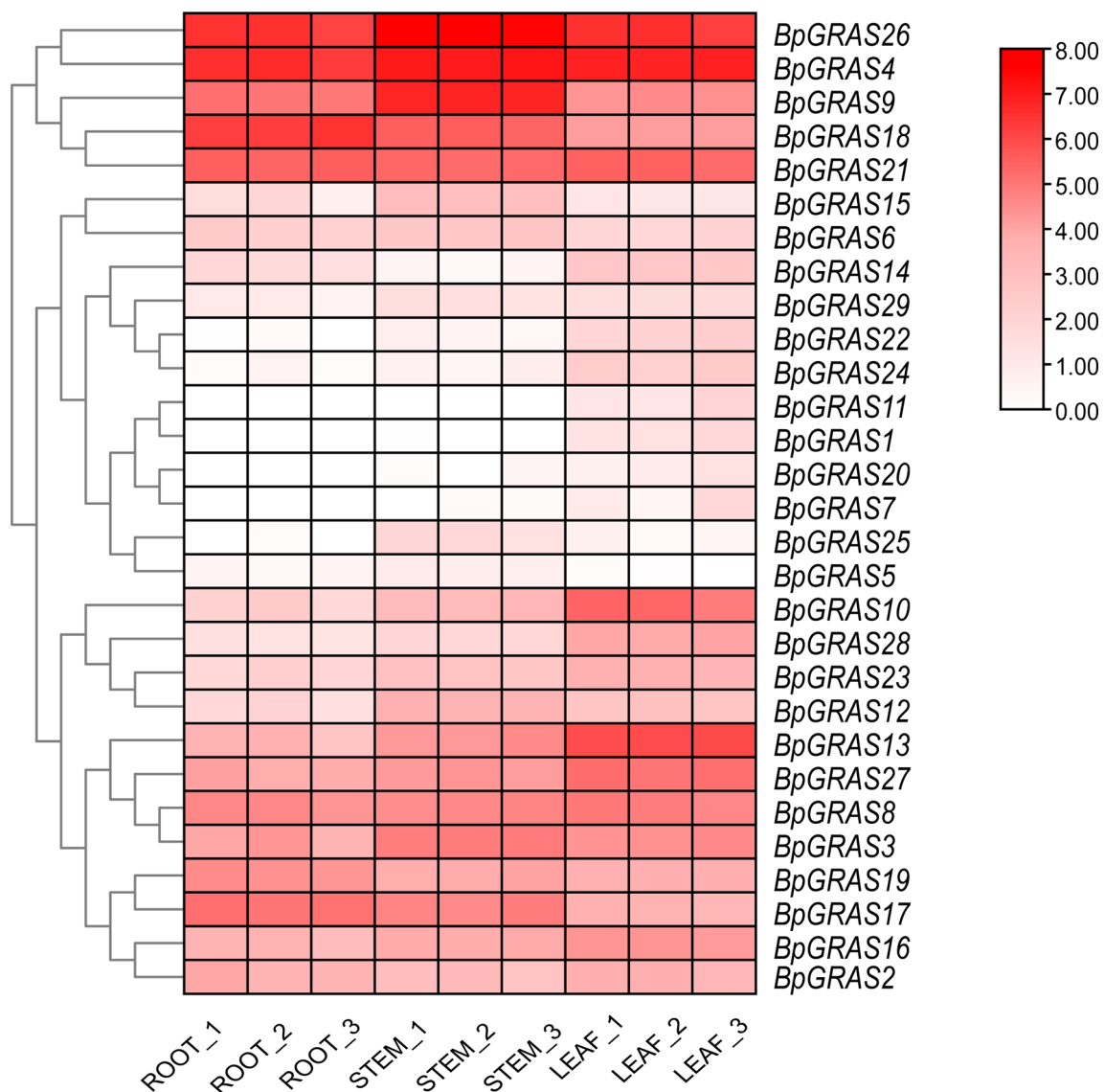
levels of *BpGRAS* genes in different tissues (Fig. 3). The heat map analysis revealed that the majority of *BpGRAS* genes were expressed across the roots, stems, and leaves of *B. purpurascens*, while notable exceptions included *BpGRAS1* and the SHR subfamily members *BpGRAS7*, *BpGRAS11*, and *BpGRAS20*. Among them, *BpGRAS1*, *BpGRAS7*, and *BpGRAS20* were not detected to be expressed in roots, and *BpGRAS11* was only detected to be expressed in leaves.

Eleven *BpGRAS* genes, *BpGRAS4* of the DELLA subfamily, *BpGRAS16* of the SCL3 subfamily, *BpGRAS17/19/27* of the LISCL subfamily, *BpGRAS3/9/18/21/26* of the PAT1 subfamily, and *BpGRAS8* of the HAM subfamily, showed high constitutive expression levels in the three tissues and might play a role in maintaining *B. purpurascens* growth and development. We also found some tissue-specifically expressed genes including *BpGRAS1*, *BpGRAS7*, *BpGRAS11*, *BpGRAS20*, *BpGRAS22*, *BpGRAS24*, and *BpGRAS28*. These genes showed significantly higher

expression levels in leaves compared to roots and stems. Only *BpGRAS25* was specifically highly expressed in stems, and its expression level was significantly higher than that in the other two tissues. The expression pattern of *BpGRAS* genes in different tissues provided a potential basis for further studies to reveal their functions.

### Expression pattern of *BpGRAS* genes under salt stress

The *B. purpurascens* leaves under salt stress and control were subjected to RNA-seq analysis. In order to observe the changes of the expression of *BpGRAS* genes after salt treatment, the TPM values of *BpGRAS* genes under salt stress and control (Table S5) were plotted to form a heat map (Fig. 4). Differential expression analysis of 29 *BpGRAS* genes in two samples was performed by DESeq2. Based on  $\log_2|\text{fold change}| \geq 1$  &  $\text{P}_{\text{adjust}} < 0.05$ , eight differentially



**Fig. 3** Expression pattern of *BpGRAS* genes in three different tissues. Each sample was repeated three times

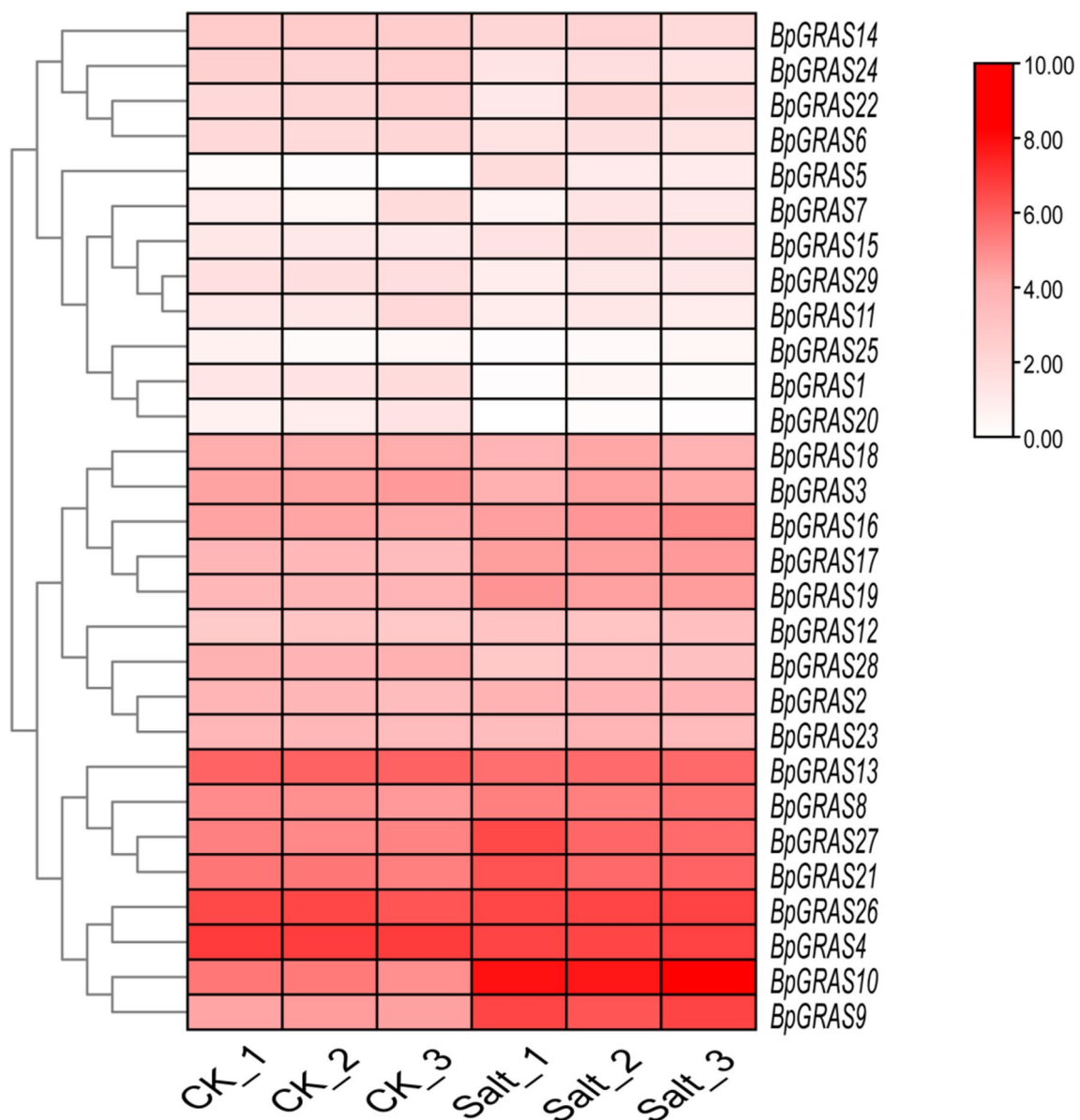
expressed *BpGRAS* genes that were identified. Six of them were up-regulated and two of them were down-regulated by salt stress. These eight *BpGRAS* genes were considered as salt-responsive genes. Four of them belonged to the *LISCL* subfamily and were up-regulated by salt stress, and the *PAT1*, *HAM* and *SCR* subfamilies contain one gene each. These salt-responsive *BpGRAS* genes might serve as candidate genes to study the specific mechanisms and adaptations of *B. purpurascens* in response to high salt environments. We selected eight *BpGRAS* genes, including four that were up-regulated by salt treatment (*BpGRAS16*, *BpGRAS12*, *BpGRAS9*, and *BpGRAS10*) and four that were down-regulated (*BpGRAS11*, *BpGRAS20*, *BpGRAS28*, and *BpGRAS1*), four of which were salt-responsive genes. The *B. purpurascens* leaves under salt stress and control were subjected to RT-qPCR. The expression trends of eight genes showed in

RT-qPCR results (Fig. 5) were completely consistent with that in RNA-seq results.

### Overexpression of *BpGRAS9* in *A. Thaliana* enhances its salt tolerance

*BpGRAS9*, a member of the *PAT1* subfamily, was significantly up-regulated in *B. purpurascens* after salt treatment. It speculated that *BpGRAS9* might play a role in *B. purpurascens* salt tolerance. In order to verify the biological function of *BpGRAS9*, transgenic *A. thaliana* with overexpressed *BpGRAS9* were constructed. Transgenic lines OE2 and OE3 with high expression levels were selected for further studies (Fig. 6A). There was no significant difference in the growth and development of transgenic and WT plants under control conditions (Fig. 6B). The root growth of transgenic and





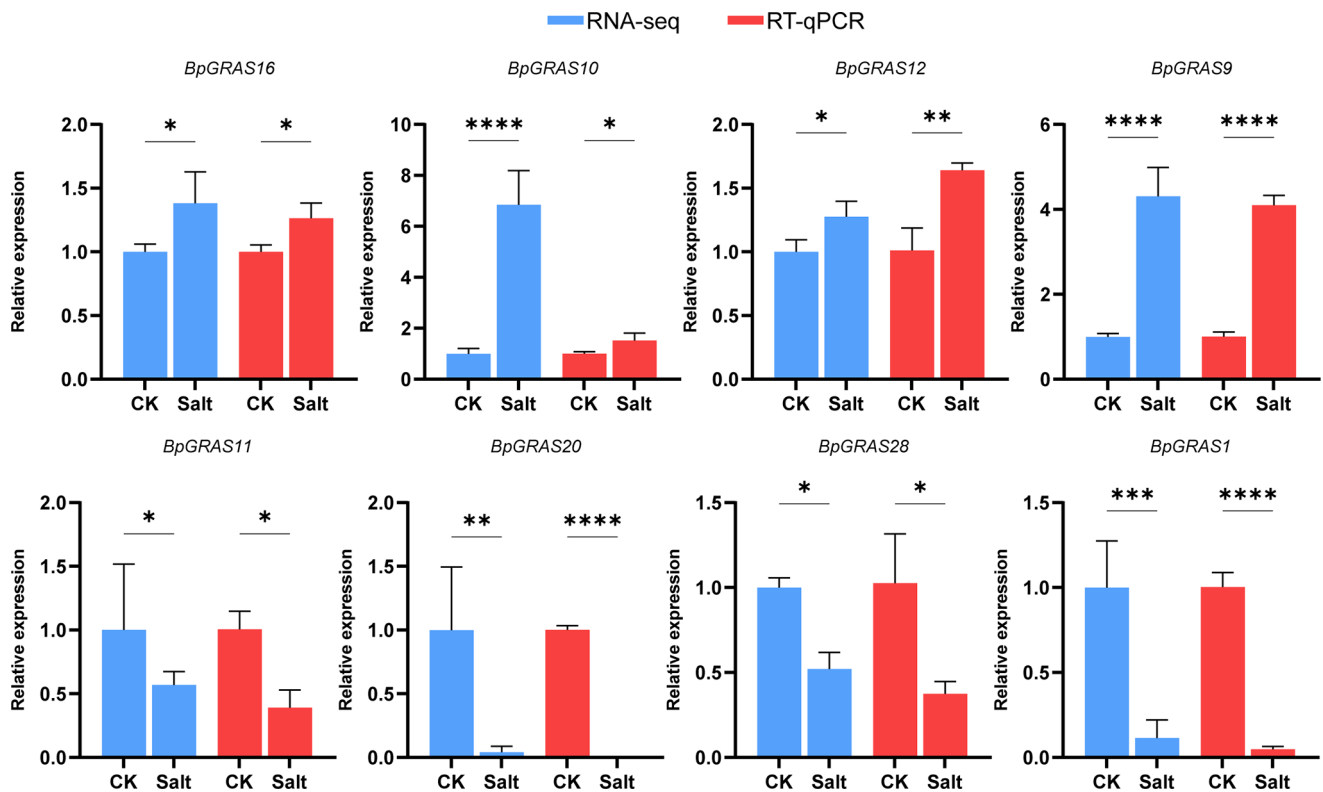
**Fig. 4** Expression patterns of *BpGRAS* genes under salt stress. Each sample was repeated three times

WT plants was significantly affected by salt under salt conditions, but the transgenic lines OE2 and OE3 had significantly higher fresh weight and root length than WT under salt conditions (Fig. 6C and D). Meanwhile, we detected some genes that have been reported to be involved in plant salt tolerance, such as *SOS1*, *SOS2*, and *SOS3* (Chen et al. 2021), in both WT and transgenic plants by RT-qPCR. We found that the expression of these genes was up-regulated in transgenic plant (Figure S1). It demonstrated that *BpGRAS9* mitigated the impact of salinity on plant growth and development, and enhancing the plant's salt tolerance.

To further substantiate the role of *BpGRAS9* in plant response to salt stress, the albino rate of transgenic and WT plants was analyzed. The results showed that the albino

rate of transgenic lines OE2 and OE3 under continuous salt stress was significantly lower than that of WT (Fig. 7A and B). Considering that salt stress can affect plant photosynthesis and thus plant growth, we examined the chlorophyll content of transgenic plants and WT under continuous salt stress. There was no significant difference between the chlorophyll contents of WT and transgenic plants under control conditions (Fig. 7C). The chlorophyll contents of both WT and transgenic plants were reduced by salt under salt stress conditions, but the chlorophyll contents of OE2 and OE3 of transgenic plants were significantly higher than those of WT (Fig. 7C). It indicated that *BpGRAS9* mitigated the impact of salinity on plant photosynthesis and reduced salt-induced damage to a certain extent.





**Fig. 5** The expression levels of the eight *BpGRAS* genes were detected by RT-qPCR under salt stress and control conditions (\* $P < 0.05$ , \*\* $P < 0.01$ , \*\*\* $P < 0.001$ , Student's t-test)

### ***BpGRAS9* reduces ROS accumulation under salt tolerance**

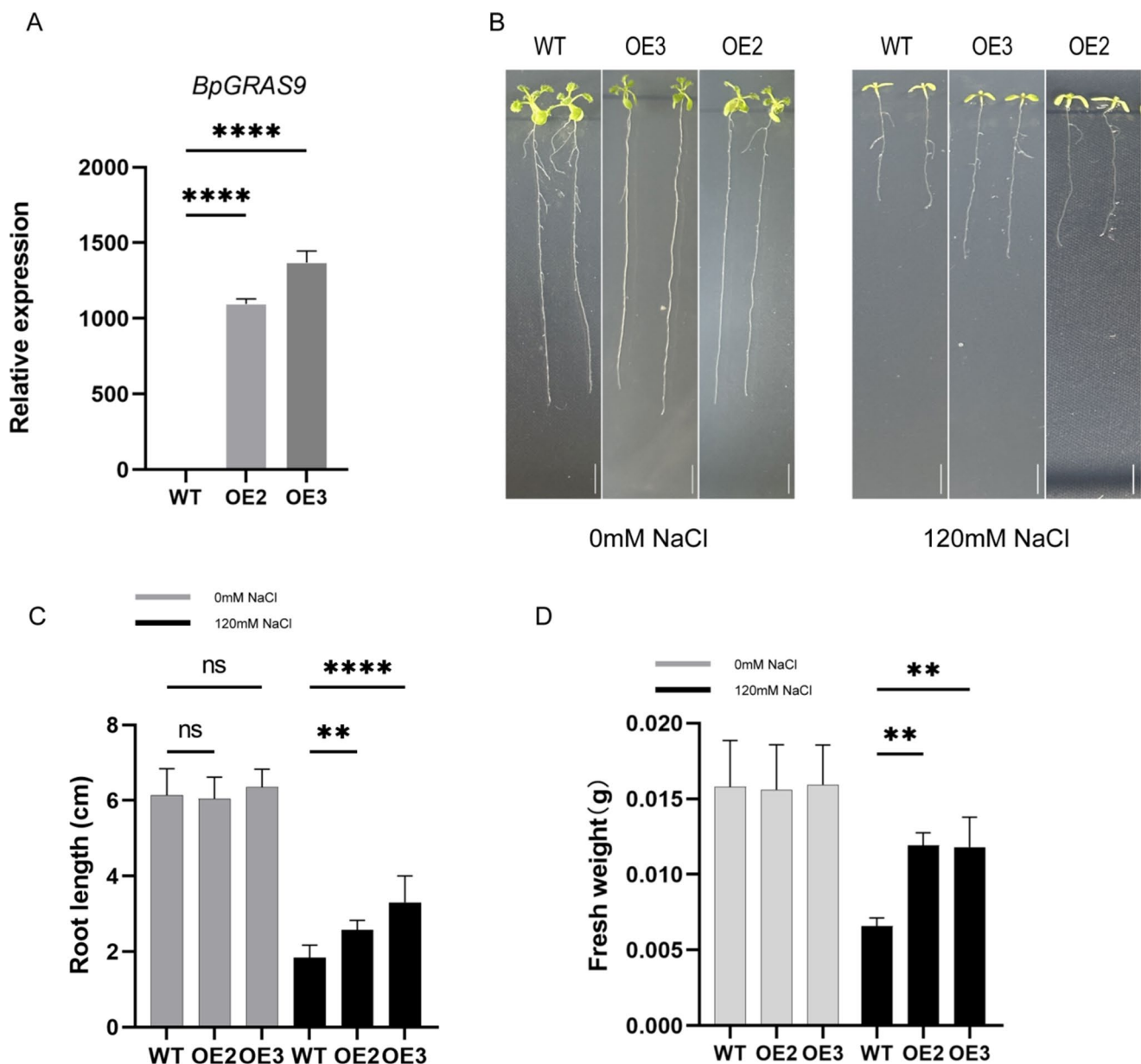
Reactive oxygen species (ROS) including  $O_2^-$ ,  $H_2O_2$ ,  $OH^-$  as well as  $O_2$  are products of oxidative metabolism, and excess ROS tend to be toxic to plants. It was found that the accumulation of ROS in plants was increased under salt tolerance (Miller et al. 2010). To observe ROS accumulation in transgenic lines OE2, OE3, and WT under salt tolerance, we stained them with nitro-blue tetrazolium (NBT) and Tetrahydrochloride (DAB) (Fig. 8A and B). As can be seen from the results, there was no significant difference in NBT and DAB staining of transgenic plants OE2, OE3 and WT under control conditions. Under salt stress, the intensity of NBT and DAB staining was increased by salt in both WT and transgenic plants, but the staining intensity of transgenic lines OE2 and OE3 was significantly lower than that of WT. It suggests that *BpGRAS9* increases salt tolerance in plants by reducing the accumulation of ROS under salt stress conditions. We further determined the levels of peroxidase (POD), catalase (CAT) and proline in the plants (Fig. 9). The results indicated that the POD, CAT and proline contents of transgenic lines OE2 and OE3 were higher than those of WT under salt conditions. It suggests that *BpGRAS9* enhances the ability of plants to be clear of

ROS by elevating the content of POD and CAT, as well as increasing the content of proline to increase the salt tolerance of plants.

### **Discussion**

GRAS transcription factors play important regulatory roles in plant growth and development and abiotic stresses (Jaiswal et al. 2022). The *GRAS* gene family in *B. purpurascens* has not been comprehensively characterized and functionally studied. In this study, we identified 29 *BpGRAS* genes and comprehensively analyzed their protein physicochemical properties, phylogenetic classification, conserved motif analysis, gene expression profiles in different tissues and under salt stress, and *BpGRAS9* was functionally characterized as an important regulator to enhance the plant's salt tolerance. The studies enhanced our comprehension of the function of *BpGRAS* and established a more foundation for exploring the molecular mechanisms underlying plant salt tolerance.

The *GRAS* genes in *B. purpurascens* were identified similar to the genome reported for *A. thaliana* (34) (Liu and Widmer 2014), and lower than rice (60) (Liu and Widmer 2014), cassava (77) (Shan et al. 2020), *Dendrobium*



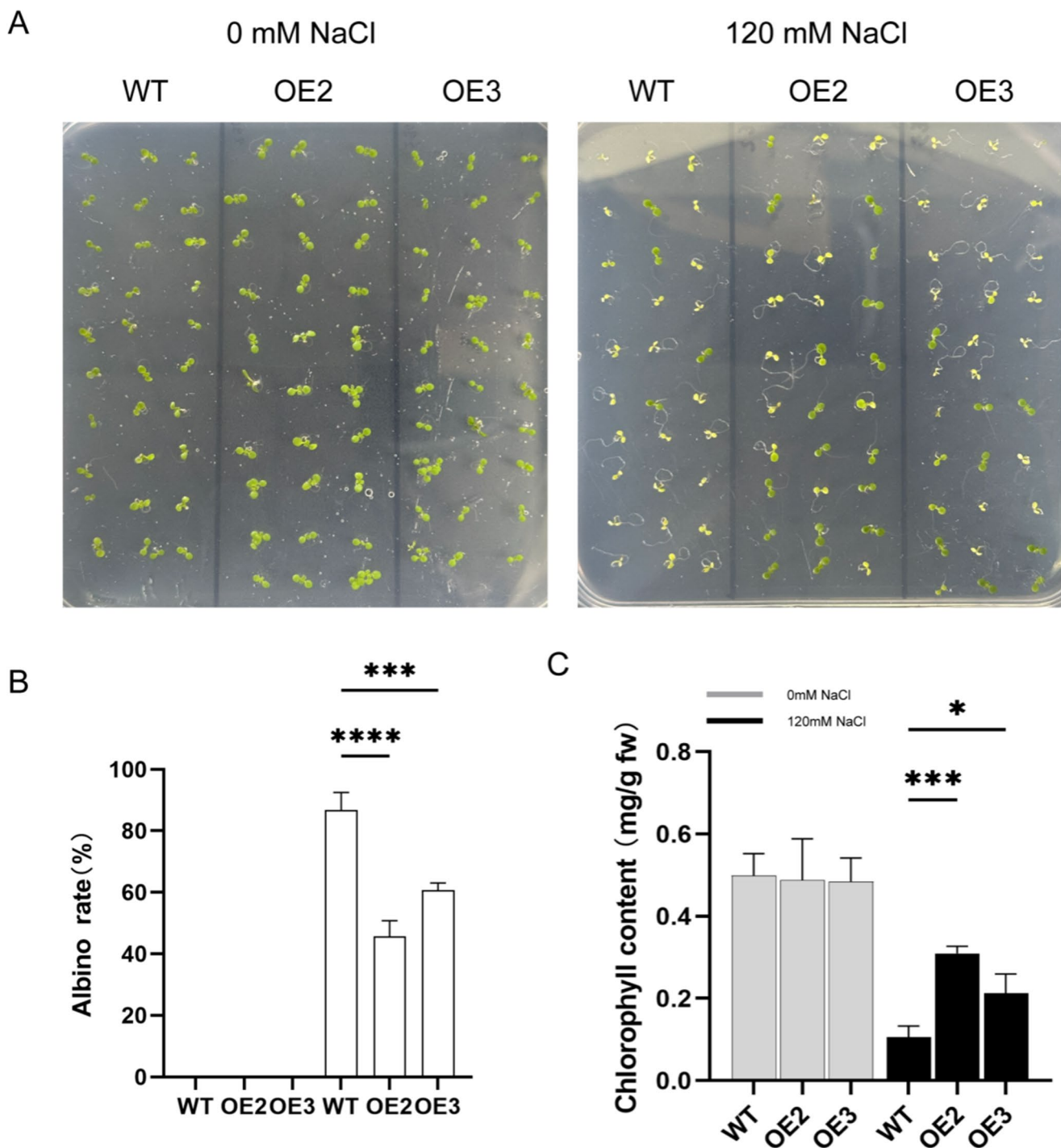
**Fig. 6** Salt stress tolerance analysis of *BpGRAS9* overexpressing (OE) *A. thaliana* seedlings. **(A)** Relative expression levels of *BpGRAS9* in transgenic Arabidopsis lines. **(B)** Phenotypes of transgenic and WT *A. thaliana* seedlings 8 days after transfer to 1/2 MS medium containing

0, 120 mM NaCl. **(C)** Root length and **(D)** fresh weight of *A. thaliana* seedlings. (\* $P < 0.05$ , \*\* $P < 0.01$ , \*\*\* $P < 0.001$ , Student's *t*-test). Bar = 0.5 cm

*chrysotoxum*(46) (Zhao et al. 2022), tobacco (53) (Chen et al. 2015a, b), grapevine (52) (Grimplet et al. 2016), soybean (117) (Wang et al. 2020), maize (86) (Sun et al. 2017), and *Medicago sativa* (Zhang et al. 2021). Variations in the number of GRAS genes across different species might result from the long-term evolutionary processes aimed at environmental adaptation. These processes likely involve distinct gene duplication events or gene/genome transfer events.

According to the *GRAS* gene family classification criteria for *(A) thaliana* and rice, 29 *BpGRAS* genes were classified

into seven subfamilies, except for *BpGRAS1*, which might belong to the *(B) purpurascens*-specific subfamily. *PAT1* is the most numerous subfamily of *BpGRAS* genes, containing seven *BpGRAS* genes. The *PAT1* subfamily played an important regulatory role in plant signalling, overexpression of *VaPAT1* increased the resistance of *(A) thaliana* to abiotic stress in the face of abiotic stresses (Yuan et al. 2015). This suggests that the *PAT1* subfamily might play a role in *(B) purpurascens* resistance responses. The *HAM*, *LISCL*, *SHR* subfamilies contained five, six, and four *BpGRAS* genes respectively, and these *BpGRAS* genes might play different

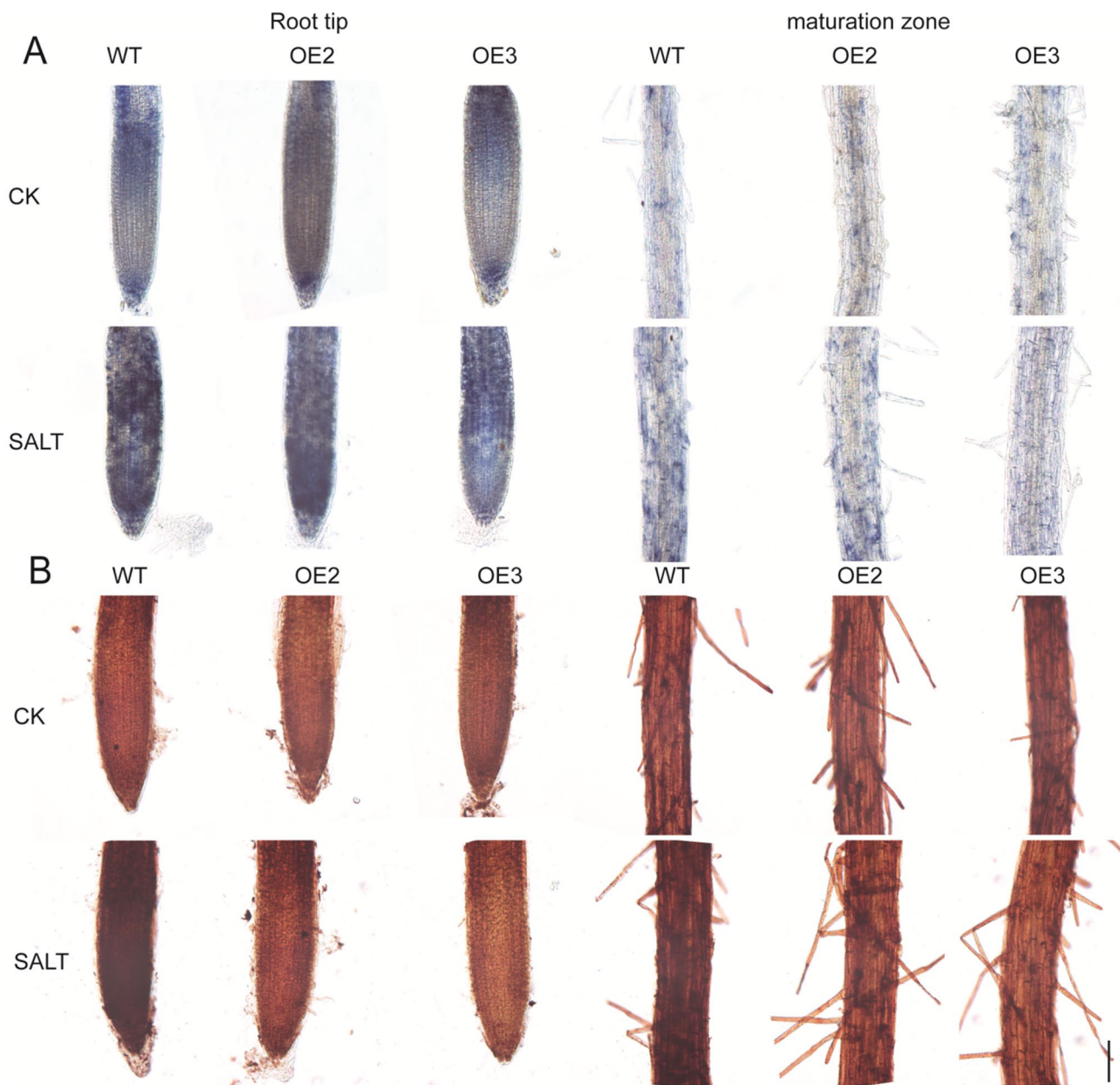


**Fig. 7** *BpGRAS9* overexpression reduced albino rate and loss of chlorophyll content in transgenic *A. thaliana*. (A) Phenotypes of transgenic and WT *A. thaliana* at 10 days of germination, (B) albino rate of WT

and transgenic *A. thaliana* in A, and (C) chlorophyll content determination of WT and transgenic *A. thaliana* in A. (\* $P < 0.05$ , \*\* $P < 0.01$ , \*\*\* $P < 0.001$ , Student's t-test)

roles in plant growth and development. The *HAM* subfamily is associated with the maintenance of apical or lateral meristematic tissues in *Petunia hybrida* plants (Jaiswal et al. 2022), the *LISCL* subfamily plays a role in the reduction of high light and salt stress damage to plants, and the *SHR/SCR* subfamilies usually form a complex to promote the

formation of endodermis to regulate the radial organisation of roots (Helariutta et al. 2000; Levesque et al. 2006). Gibberellins (GA) are endogenous regulators of plant growth and development (Hedden and Phillips 2000). The *DELLA* subfamily encodes inhibitors in GA signalling, and (*A. thaliana* *SCL3* is specifically expressed in the endodermis



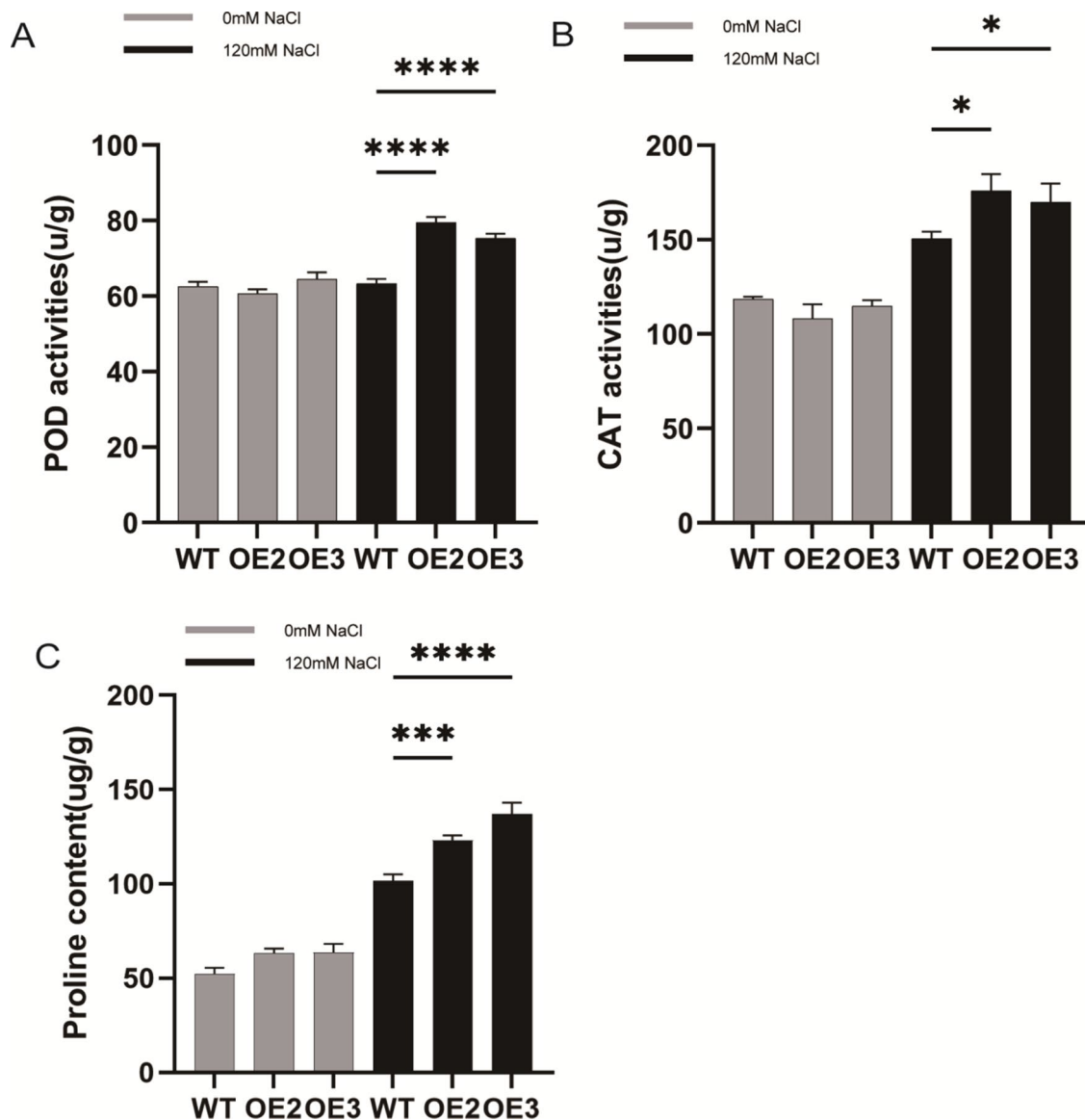
**Fig. 8** BpGRAS9 reduces ROS accumulation under salt conditions. (**A**) NBT staining of WT and transgenic lines OE2 and OE3 under control and salt conditions, (**B**) DAB staining of WT and transgenic lines OE2 and OE3 under control and salt conditions. Bar = 100  $\mu$ m

of roots and acts antagonistically to *DELLA* in the GA signalling pathway, positively regulating the GA pathway (Heo et al. 2011; Yoshida et al. 2014). The rice-specific *OS19* subfamily was not found in the *GRAS* gene family of (*B. purpurascens*, suggesting that *B. purpurascens* is closer to *A. thaliana* in its evolutionary development.

The structure of GRAS protein is very typical in that its N-terminal intrinsically disordered region can bind to specific proteins and thus specifically function in plant growth and development, and the C-terminal usually contains several very conserved structural domains (Sun et al.

2011, 2012). Among the C-terminal structural domains of GRAS proteins, the VHIIID structural domain is the most conserved, and in this study the results of gene sequence comparison showed that all BpGRAS members contain the VHIIID structural domain, and the results can be seen that almost all the members (except the SCR subfamily) had amino acid substitutions of varying degrees, but histidine (H) and aspartic acid (D) are extremely stable (Fig. 10). All members of the DELLA subfamily have isoleucine (I) replaced by valine(V), and PAT1 also has some members with isoleucine (I) replaced by valine(V), along with other





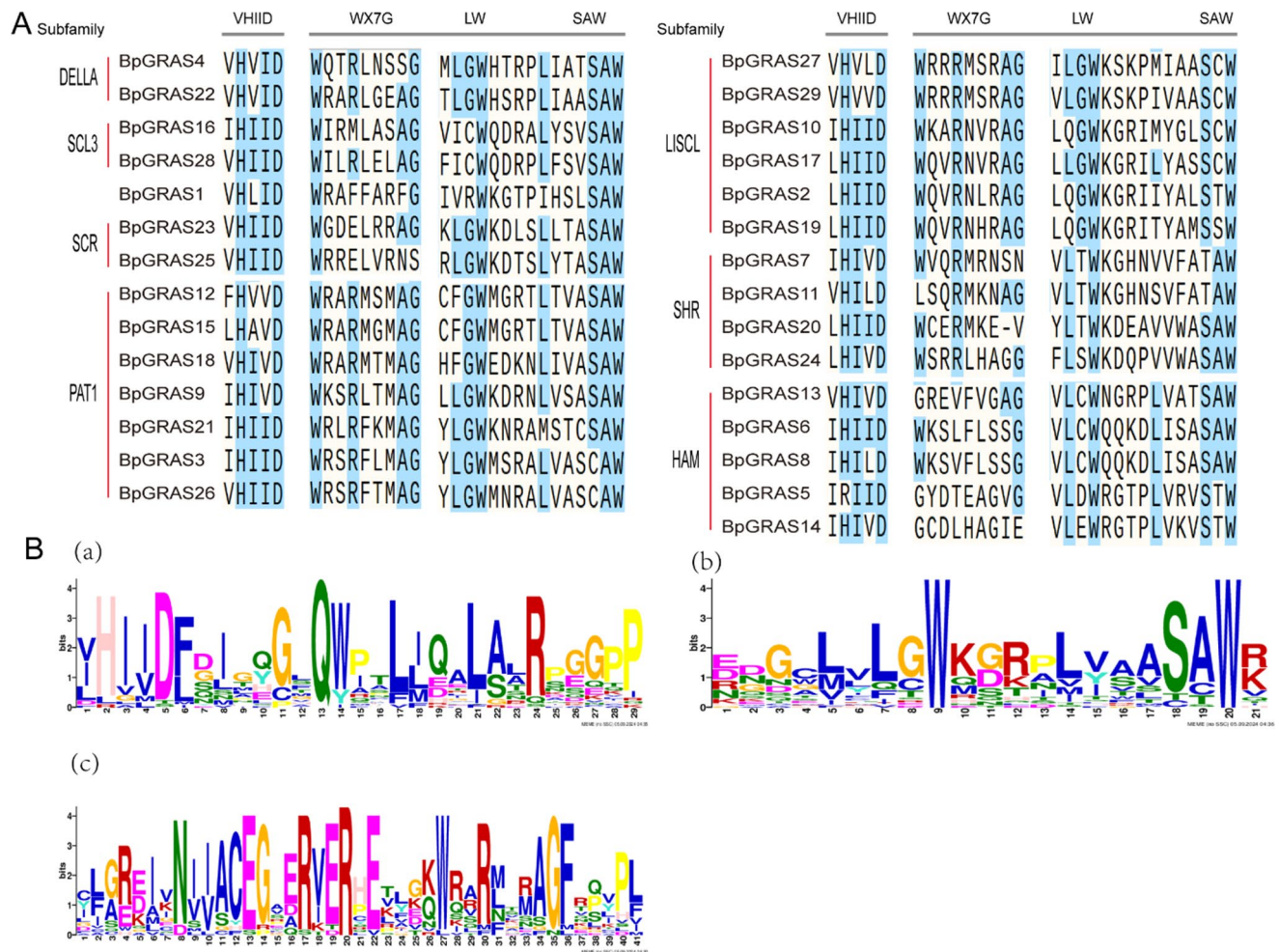
**Fig. 9** POD, CAT and proline content of transgenic plants and WT. (A) POD content of WT and transgenic lines OE2 and OE3 under control and salt conditions, (B) CAT content of WT and transgenic lines OE2

and OE3 under control and salt conditions, (C) Proline content of WT and transgenic lines OE2 and OE3 under control and salt conditions. (\* $P < 0.05$ , \*\* $P < 0.01$ , \*\*\* $P < 0.001$ , Student's t-test)

amino acid substitutions. It has been demonstrated that the PAT1 and DELLA subfamilies are involved in plant light signalling and GA signalling (Bolle et al. 2000; Bouré and Arnaud 2023), and also play a role in stress response (Yuan et al. 2015). The occurrence of amino acid substitutions in the PAT1 and DELLA subfamilies of *B. purpurascens* might be a result of adaptation to the environment during the evolutionary process. In addition, the SAW structural domain is also a conserved structural domain at the C-terminal, which consists of three units, WX7G, LW, and SAW. Most of BpGRAS proteins in this study contained this motif, and the exact function of the SAW motifs is unclear in plants. Mutations in these motifs in NSP1 affect plant nodulation

(Heckmann et al. 2006). They might be intrinsically associated as conserved motifs essential for maintaining the structural and functional integrity of GRAS proteins.

The expression levels of the *BpGRAS* genes in different tissues and the changes in the expression profiles after abiotic stress can better infer the function of the genes, and the expression levels of *B. purpurascens* BpGRAS9 were high in three tissues of roots, stems, and leaves, and the expression levels were increased significantly after treatment with salt stress, which can be deduced to be involved in the salt-tolerance response in *B. purpurascens*. This study showed that the root length and weight of fresh-weight of BpGRAS9-transgenic plants were significantly higher than those of WT



**Fig. 10** VHIID and SAW motifs in the amino acid sequence of GRAS protein. (A) BpGRAS protein sequence alignment results, (B) sequence identification of motifs, (a) sequence of the VHIID structural domain, (b) and (c) sequence of the SAW structural domain

in salty conditions, demonstrating that *BpGRAS9* enhanced the plant salt tolerance. Meanwhile, the occurrence of albinism in *BpGRAS9*-transgenic plants cultivated under saline conditions was markedly lower compared to WT plants. Additionally, the chlorophyll content in these transgenic plants was significantly higher than that observed in WT plants. *BpGRAS9* reduced ROS accumulation under salt stress and increased proline content. These results suggest that *BpGRAS9* can mitigate the adverse effects of salt stress on plant photosynthesis, as well as improve the ability of plants to clear ROS under salt stress, thereby promoting plant growth in salt environments. Salinity is now a common adversity that threatens the growth and development of plants. Salt stress can seriously harm the normal growth of plants and inhibit crop yields (Arzani 2008). Rice and maize yields are severely reduced under salt stress, and the yields of crops such as potatoes, aubergines, peppers, and lettuce are significantly affected by salt stress (Ngara et al. 2012).

This study provides a new basis for the improvement of salt tolerance breeding of crops.

**Supplementary Information** The online version contains supplementary material available at <https://doi.org/10.1007/s11103-025-01562-0>.

**Acknowledgements** This work was supported by grants from the National Natural Science Foundation of China (No. 31900164), the Provincial Natural Science Foundation of Sichuan (No. 2022NSFSC1612 and 2024NSFSC0384) and the Fundamental Research Funds for the Central Universities (No. 2682023ZTPY073).

**Author contributions** Qiankun Zhu designed the project. Xin Lyu, Nuomei Xu, Jie Chen, Wenqing Wang, and Feiyang Yan ran the experiments. Xin Lyu and Nuomei Xu analyzed the results. Xin Lyu and Qiankun Zhu wrote the manuscript.

## Declarations

**Conflict of interest** We declare that all authors have no conflict of interest.

## References

- Arzani A (2008) Improving salinity tolerance in crop plants: a biotechnological view. *Vitro Cell Dev Biology-Plant* 44:373–383
- Bolle C (2004) The role of GRAS proteins in plant signal transduction and development. *Planta* 218:683–692
- Bolle C, Koncz C, Chua NH (2000) PAT1, a new member of the GRAS family, is involved in phytochrome A signal transduction. *Genes Dev* 14:1269–1278
- Bouré N, Arnaud N (2023) Molecular GA pathways as conserved integrators for adaptive responses. *Plant Biol* 25:649–660
- Chen K, Li H, Chen Y, Zheng Q, Li B, Li Z (2015a) TaSCL14, a Novel Wheat (*Triticum aestivum* L.) GRAS Gene, regulates Plant Growth, Photosynthesis, Tolerance to photooxidative stress, and Senescence. *J Genet Genomics* 42:21–32
- Chen YQ, Tai SS, Wang DW, Ding AM, Sun TT, Wang WF, Sun YH (2015b) Homology-based analysis of the GRAS gene family in tobacco. *Genet Mol Res* 14:15188–15200
- Chen CJ, Chen H, Zhang Y, Thomas HR, Frank MH, He YH, Xia R (2020a) TBtools: an integrative Toolkit developed for interactive analyses of big Biological Data. *Mol Plant* 13:1194–1202
- Chen M, Ye C, Zhu J, Zhang P, Jiang Y, Lu X, Wu H (2020b) Bergenin as a Novel Urate-lowering therapeutic strategy for Hyperuricemia. *Front Cell Dev Biology* 8
- Chen XX, Ding YL, Yang YQ, Song CP, Wang BS, Yang SH, Guo Y, Gong ZZ (2021) Protein kinases in plant responses to drought, salt, and cold stress. *J Integr Plant Biol* 63:53–78
- Clough SJ, Bent AF (1998) Floral dip: a simplified method for *Agrobacterium*-mediated transformation of *Arabidopsis thaliana*. *Plant J* 16:735–743
- Day RB, Shibuya N, Minami E (2003) Identification and characterization of two new members of the GRAS gene family in rice responsive to *N*-acetylchitooligosaccharide elicitor. *Biochimica et Biophysica Acta (BBA) - gene structure and expression*. 1625:261–268
- Day RB, Tanabe S, Koshioka M, Mitsui T, Itoh H, Ueguchi-Tanaka M, Matsuoka M, Kaku H, Shibuya N, Minami E (2004) Two rice GRAS family genes responsive to *N*-acetylchitooligosaccharide elicitor are induced by phytoactive gibberellins: evidence for cross-talk between elicitor and gibberellin signaling in rice cells. *Plant Mol Biol* 54:261–272
- Edgar RC (2004) MUSCLE: multiple sequence alignment with high accuracy and high throughput. *Nucleic Acids Res* 32:1792–1797
- Greb T, Clarenz O, Schäfer E, Müller D, Herrero R, Schmitz G, Theres K (2003) Molecular analysis of the *LATERAL SUPPRESSOR* gene in *Arabidopsis* reveals a conserved control mechanism for axillary meristem formation. *Genes Dev* 17:1175–1187
- Grimplet J, Agudelo-Romero P, Teixeira RT, Martinez-Zapater JM, Fortes AM (2016) Structural and functional analysis of the GRAS Gene Family in Grapevine indicates a role of GRAS proteins in the Control of Development and Stress Responses. *Front Plant Sci* 7:22
- Hakoshima T (2018) Structural basis of the specific interactions of GRAS family proteins. *FEBS Lett* 592:489–501
- Heckmann AB, Lombardo F, Miwa H, Perry JA, Bunnewell S, Parniske M, Wang TL, Downie JA (2006) *Japonicusonicus* Nodulrequiesqtwoes TwodomainDregulatorslaones, Owhich isifunctionallyoconservedservedNon-legumeLegume. *Plant Physiol* 142:1739–1750
- Hedden P, Phillips AL (2000) Gibberellin metabolism: new insights revealed by the genes. *Trends Plant Sci* 5:523–530
- Helariutta Y, Fukaki H, Wysocka-Diller J, Nakajima K, Jung J, Sena G, Hauser MT, Benfey PN (2000) The *SHORT-ROOT* Gene controls Radial patterning of the *Arabidopsis* Root through Radial Signaling. *Cell* 101:555–567
- Heo J-O, Chang KS, Kim IA, Lee M-H, Lee SA, Song S-K, Lee MM, Lim J (2011) Funneling of gibberellin signaling by the GRAS transcription regulator SCARECROW-LIKE 3 in the *Arabidopsis* root. *Proceedings of the National Academy of Sciences* 108:2166–2171
- Jaiswal V, Kakkar M, Kumari P, Zinta G, Gahlaut V, Kumar S (2022) Multifaceted roles of GRAS transcription factors in growth and stress responses in plants. *iScience* 25
- Jiang CF, Belfield EJ, Cao Y, Smith JAC, Harberd NP (2013) An *Arabidopsis* Soil-Salinity-Tolerance Mutation confers Ethylene-mediated enhancement of Sodium/Potassium Homeostasis. *Plant Cell* 25:3535–3552
- Koizumi K, Gallagher KL (2013) Identification of SHRUBBY, a SHORT-ROOT and SCARECROW interacting protein that controls root growth and radial patterning. *Development* 140:1292–1300
- Koul B, Kumar A, Yadav D, Jin J-O (2020) *Bergenia* Genus: traditional uses, Phytochemistry and Pharmacology. *Molecules* 25
- Lee M-H, Kim B, Song S-K, Heo J-O, Yu N-I, Lee SA, Kim M, Kim DG, Sohn SO, Lim CE, Chang KS, Lee MM, Lim J (2008) Large-scale analysis of the GRAS gene family in *Arabidopsis thaliana*. *Plant Mol Biol* 67:659–670
- Levesque MP, Vernoux T, Busch W, Cui HC, Wang JY, Blilou I, Hassan H, Nakajima K, Matsumoto N, Lohmann JU, Scheres B, Benfey PN (2006) Whole-genome analysis of the SHORT-ROOT developmental pathway in *Arabidopsis*. *Plos Biology* 4:1284–1284
- Li B, Dewey CN (2011) RSEM: accurate transcript quantification from RNA-Seq data with or without a reference genome. *BMC Bioinformatics* 12:16
- Liu X, Widmer A (2014) Genome-wide comparative analysis of the GRAS Gene Family in *Populus*, *Arabidopsis* and Rice. *Plant Mol Biology Report* 32:1129–1145
- Liu Y, An ZX, He YL (2023) The traditional uses, phytochemistry, pharmacology and toxicology of *Bergenia purpureascens*: a review comments and suggestions. *Heliyon* 9
- Liu Y-Y, Zhang Y, Jiang L, Lu Q-Y, Ye R-H, Guo Z-Y, Zhao Y-L, Luo X-D (2024) The whitening effect of cuscutein responsible for traditional use of *Bergenia purpureascens*. *J Ethnopharmacol* 326
- Miller G, Suzuki N, Ciftci-Yilmaz S, Mittler R (2010) Reactive oxygen species homeostasis and signalling during drought and salinity stresses. *Plant Cell Environ* 33:453–467
- Morohashi K, Minami M, Takase H, Hotta Y, Hiratsuka K (2003) Isolation and characterization of a Novel GRAS Gene that regulates meiosis-associated gene expression. *J Biol Chem* 278:20865–20873
- Myouga F, Hosoda C, Umezawa T, Iizumi H, Kuromori T, Motohashi R, Shono Y, Nagata N, Ikeuchi M, Shinozaki K (2008) A heterocomplex of Iron Superoxide dismutases defends Chloroplast Nucleoids against oxidative stress and is essential for Chloroplast Development in *Arabidopsis*. *Plant Cell* 20:3148–3162
- Ngara R, Ndimba R, Borch-Jensen S, Jensen ON, Ndimba B (2012) Identification and profiling of salinity stress-responsive proteins in *Sorghum bicolor* seedlings. *J Proteom* 75:4139–4150
- Pysh LD, Wysocka-Diller JW, Camilleri C, Bouchez D, Benfey PN (2002) The GRAS gene family in *Arabidopsis*: sequence characterization and basic expression analysis of the SCARECROW-LIKE genes. *Plant J* 18:111–119
- Schumacher K, Schmitt T, Rossberg M, Schmitz C, Theres K (1999) The *lateral suppressor (ls)* gene of tomato encodes a new member of the VHIID protein family. *Proc Natl Acad Sci USA* 96:290–295
- Shan Z, Luo X, Wu M, Wei L, Fan Z, Zhu Y (2020) Genome-wide identification and expression of GRAS gene family members in cassava. *BMC Plant Biol* 20
- Stuurman J, Jäggi F, Kuhlemeier C (2002) Shoot meristem maintenance is controlled by a GRAS-gene mediated signal from differentiating cells. *Genes Dev* 16:2213–2218

- Sun XL, Xue B, Jones WT, Rikkerink E, Dunker AK, Uversky VN (2011) A functionally required unfoldome from the plant kingdom: intrinsically disordered N-terminal domains of GRAS proteins are involved in molecular recognition during plant development. *Plant Mol Biol* 77:205–223
- Sun XL, Jones WT, Rikkerink EHA (2012) GRAS proteins: the versatile roles of intrinsically disordered proteins in plant signalling. *Biochem J* 442:1–12
- Sun M-x, Guo Y, Wu H, Li X, Li Q, Zhao X, Duan X, An Y, Lv W, An H (2017) Identification and expression of *GRAS* family genes in maize (*Zea mays* L). *PLoS ONE* 12
- Tian C (2004) Genome-wide comparative analysis of the GRAS Gene Family in *Populus*, *Arabidopsis* and *Rice*. *Plant Mol Biology Report* 54:519–532
- Tian CG, Wan P, Sun SH, Li JY, Chen MS (2004) Genome-wide analysis of the GRAS gene family in rice and *Arabidopsis*. *Plant Mol Biol* 54:519–532
- Torres-Galea P, Huang L-F, Chua N-H, Bolle C (2006) The GRAS protein SCL13 is a positive regulator of phytochrome-dependent red light signaling, but can also modulate phytochrome A responses. *Mol Genet Genomics* 276:13–30
- Torres-Galea P, Hirtreiter B, Bolle C, Two GRAS, Proteins (2013) SCARECROW-LIKE21 and PHYTOCHROME a SIGNAL TRANSDUCTION1, function cooperatively in Phytochrome A Signal Transduction. *Plant Physiol* 161:291–304
- Wang T-T, Yu T-F, Fu J-D, Su H-G, Chen J, Zhou Y-B, Chen M, Guo J, Ma Y-Z, Wei W-L, Xu Z-S (2020) Genome-wide analysis of the GRAS Gene Family and Functional Identification of *GmGRAS37* in Drought and Salt Tolerance. *Front Plant Sci* 11
- Wei Y, Zhang Y, Li Q, Shen K, Zhou J (2021) Identification of DNA barcode of medicinal plants containing bergenin. *West China J Pharm Sci* 36:44–50
- Wen K, Li XL, Yin T, Chen CY, Zhu L, Zi YQ, Zhao K, Zhou XY, Liu XZ, Zhang HY (2024) Genome-wide identification of carotenoid cleavage oxygenase genes in Orah mandarin and the mechanism by which *CrCCD4b1* affects peel color. *Sci Hort* 338
- Yan R, Xie B, Xie K, Liu Q, Sui S, Wang S, Chen D, Liu J, Chen R, Dai J, Yang L (2024) Unravelling and reconstructing the biosynthetic pathway of bergenin. *Nat Commun* 15
- Yin D, Zhao Z, Guo F, Zhou G, Wang S, Li J (2017) Screening of reference genes for real-time quantitative PCR in *Bergenia purpurascens*. *Genomics Appl Biology* 36:4256–4262
- Yin T, Xu R, Zhu L, Yang XY, Zhang MJ, Li XL, Zi YQ, Wen K, Zhao K, Cai HB, Liu XZ, Zhang HY (2024) Comparative analysis of the *PAL* gene family in nine citrus provides new insights into the stress resistance mechanism of *Citrus species*. *BMC Genomics* 25
- Yoshida H, Hirano K, Sato T, Mitsuda N, Nomoto M, Maeo K, Koketsu E, Mitani R, Kawamura M, Ishiguro S, Tada Y, Ohme-Takagi M, Matsuoka M, Ueguchi-Tanaka M (2014) DELLA protein functions as a transcriptional activator through the DNA binding of the INDETERMINATE DOMAIN family proteins. *Proc Natl Acad Sci USA* 111:7861–7866
- Yuan Y, Fang L, Karungo SK, Zhang L, Gao Y, Li S, Xin H (2015) Overexpression of *VaPAT1*, a GRAS transcription factor from *Vitis amurens*, confers abiotic stress tolerance in *Arabidopsis*. *Plant Cell Rep* 35:655–666
- Zhang S-S, Liao Z-X, Huang R-Z, Gong C-C, Ji L-J, Sun H-F (2017) A new aromatic glycoside and its anti-proliferative activities from the leaves of *Bergenia purpurascens*. *Nat Prod Res* 32:668–675
- Zhang H, Liu XQ, Wang XM, Sun M, Song R, Mao PS, Jia SG (2021) Genome-wide identification of GRAS Gene Family and their responses to abiotic stress in *Medicago sativa*. *Int J Mol Sci* 22:21
- Zhang X, Yu F, Lyu X, Chen J, Zeng H, Xu N, Wu Y, Zhu Q (2023) Transcriptome profiling of *Bergenia purpurascens* under cold stress. *BMC Genomics* 24
- Zhang MY, Lu PP, Zheng YT, Huang X, Liu JN, Yan H, Quan HG, Tan R, Ren FM, Jiang HZ, Zhou JY, Liao H (2024) Genome-wide identification of *AP2/ERF* gene family in *Coptis Chinensis* Franch reveals its role in tissue-specific accumulation of benzylisoquinoline alkaloids. *BMC Genomics* 25
- Zhao X, Liu D-K, Wang Q-Q, Ke S, Li Y, Zhang D, Zheng Q, Zhang C, Liu Z-J, Lan S (2022) Genome-wide identification and expression analysis of the GRAS gene family in *Dendrobium Chrysotoxum*. *Front Plant Sci* 13
- Zhou HP, Shi HF, Yang YQ, Feng XX, Chen X, Xiao F, Lin HH, Guo Y (2023) Insights into plant salt stress signaling and tolerance. *J Genet Genomics* 51:16–34
- Zhu Q, Wu Y, Zhang X, Xu N, Chen J, Lyu X, Zeng H, Yu F (2024) Metabolomic and transcriptomic analyses reveals candidate genes and pathways involved in secondary metabolism in *Bergenia purpurascens*. *BMC Genomics* 25:1083
- Zi YQ, Zhang MJ, Yang XY, Zhao K, Yin T, Wen K, Li XL, Liu XZ, Zhang HY (2024) Identification of the sweet orange (*Citrus sinensis*) *bHLH* gene family and the role of *CsbHLH55* and *CsbHLH87* in regulating salt stress. *Plant Genome* 17

**Publisher's note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.